

Poverty and Inequality Platform (PIP): an introduction using Stata

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Outline

1. Introduction to the Poverty and Inequality Platform (PIP)
2. PIP user interface
3. Accessing the PIP API with Stata
4. Exercises
5. Additional resources

1. Introduction

Poverty data at the World Bank

- The World Bank mission is to end extreme poverty and boost shared poverty on a livable planet.
- To that end, the World Bank often helps countries field household surveys to track welfare.
- Some countries do not receive assistance yet the Bank is granted access to the data through collaborations with the National Statistical Office.
- For high-income countries, poverty data is sourced from the Luxembourg Income Study or EU-SILC.

The Poverty and Inequality Platform (PIP)

- PIP is the source of poverty and inequality data for the World Bank.
- PIP is the source of data for several targets of the Sustainable Development Goals on ending poverty (SDG1) and reducing inequality (SDG10).
- PIP is the joint collaboration between different parts of the World Bank through the Global Poverty Working Group
 - Development Data Group
 - Development Research Group
 - Poverty and Equity Global Department

PIP motivation

- **Scope:** PIP is the World Bank's data portal for poverty, prosperity, and inequality estimates.
- **Accessibility:** PIP data is available through Application Programming Interface (API)
 - User Interface (UI) – for quick data access and visualizations
 - Stata and R wrappers – for advanced users, researchers, students
 - Microdata access (for some surveys) through Microdata Online.
- **Open data, open code, reproducible research:**
 - PIP code is publicly available, and the PIP Methodology Handbook provides details on the poverty, prosperity, and inequality estimates reported in PIP.
 - PIP typically is updated twice a year - Spring/Fall. One can, however, access various vintages of the data.

PIP: Transparency

All code necessary to create the poverty estimates are publicly available (<https://github.com/PIP-Technical-Team>)

master pipaux / R / pip_gdp.R

randrescastaneda [replace maindir = getOption\("pipaux.maindir"\) for gls\\$PI...](#) Latest comm

2 contributors

34 lines (31 sloc) | 936 Bytes

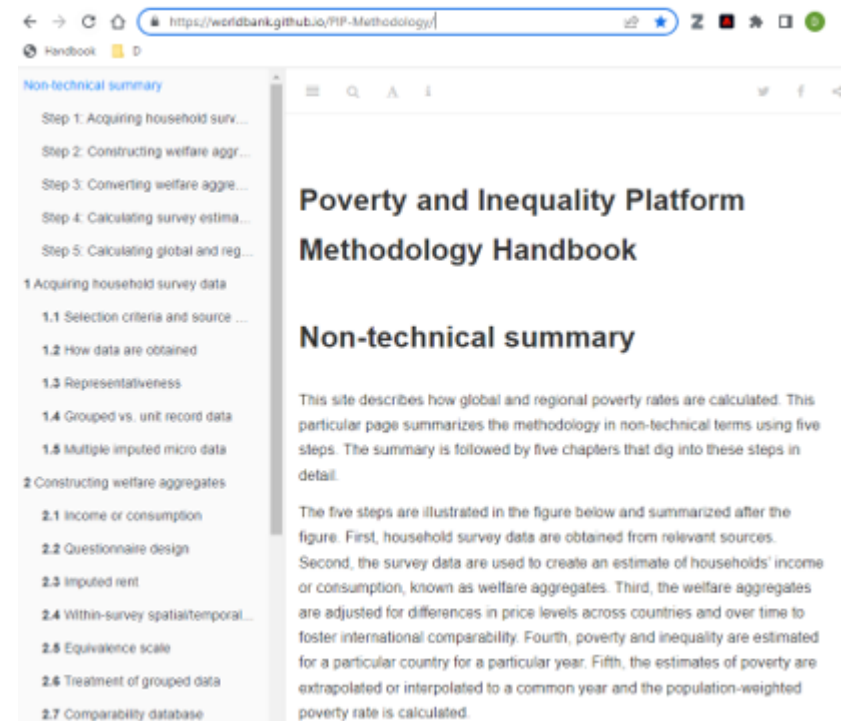
```

1 #' PIP GDP
2 #'
3 #' Update or load GDP data.
4 #'
5 #' @inheritParams pip_prices
6 #' @param maddison_action character: Either "load" or "update". Default is "update".
7 #' @param weo_action character: Either "load" or "update". Default is "update".
8 #' @export
9 pip_gdp <- function(action = "update",
10                     maddison_action = "load",
11                     weo_action = "load",
12                     force = FALSE,
13                     maindir = gls$PIP_DATA_DIR) {
14   measure <- "gdp"
15
16   # Update Maddison Project Data
17   if (maddison_action == "update") {
18     pip_maddison(force = force, maindir = maindir)
19   }
20
21   # Update WEO data
22   if (weo_action == "update") {
23     pip_gdp_weo(force = force, maindir = maindir)

```

PIP: Transparency

A description of the entire methodology is available in a technical handbook
(<https://worldbank.github.io/PIP-Methodology/>)



Data release notes and other publications

Publications are available detailing the updates in each release, methodological assumptions, and other technical improvements
(<https://worldbank.github.io/publication/>)



Pandemic, prices, and poverty

DANIEL GERSZON MAHLER, NISHANT YONZAN, RUTH HILL, CHRISTOPH LAKNER, HAORYU WU & NOBUO YOSHIDA | APRIL 13, 2022

[More complete list of publications with PIP data](#)

Welfare measure reported in PIP



Welfare measure reported in PIP

Step 1

Step 2

Step 3

Step 4

Step 5

Acquiring Household Data

- Sources:
 - LIC: Surveys collected by National Statistical Offices
 - HIC:
 - Europe: EU-SILC
 - Other (and earlier years): Luxembourg Income Study (LIS)
- Data input type: welfare (consumption/income) data at the household or group level

Welfare measure reported in PIP



- **Sources:**
 - LIC: Surveys collected by National Statistical Offices
 - HIC:
 - Europe: EU-SILC.
 - Other (and earlier years): Luxemburg Income Study (LIS)
- Data input type: welfare (consumption/income) data at the household or group level.

Welfare measure reported in PIP



- **Income vs Consumption** (`welfare_type`) -> consumption preferred.
- Zeros dropped.
- **Micro or Grouped Data** (`distribution_type`)
- **Additional comparability considerations:**
 - Questionnaire design changes.
 - Equivalence scales (-> not used)
 - Spatio/temporal adjustments.
 - Imputed rent (-> excluded)
 - Tracked in PIP: `comparable_spell` and `survey_comparability`.

Welfare measure reported in PIP



- Consumer Price Indices (CPIs) (`cpi`) & Purchasing Power Parities (PPPs) (`ppp`)
- Derivation of the International Poverty Line (`poverty_line`) which is \$3.00/day in 2021 PPPs and is based on the median of harmonized national poverty lines from the poorest countries.
- Derivation of higher poverty lines which are \$4.20 and \$8.30 in 2021 PPPs and are based on median values of national poverty lines of lower- and upper-middle income countries, respectively.
- Derivation of societal poverty line (`spl`)

Welfare measure reported in PIP



- Poverty and Inequality metrics are calculated using adjusted welfare aggregates.
- All methodology is available in the PIP Methodological Handbook on the website
- Source code is available on github through the package `wbpip`. Check the methodological handbook for links to the specific functions.

Welfare measure reported in PIP



Few steps:

1. Extrapolations and/or Interpolations (`fill_gaps`)
2. Nowcasts (`nowcast`)
3. Regional classification
4. Population adjustments
5. **Coverage rules** applied.

2. Accessing PIP through the user interface

3. Accessing PIP in Stata

Installing the Stata pip.ado

We recommend installing the package through GitHub by first running the following, which allows you to install Stata packages from GitHub:

- `net install github, from("https://haghigh.github.io/github/")`
- You can now install the Stata package by running
- `github install worldbank/pip`

Installing the Stata pip.ado

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- `net install github, from("https://haghish.github.io/github/")`
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- `github install worldbank/pip`

More information about the package:

<https://github.com/worldbank/pip/>

Two main functions

- **pip cl:** country estimates of poverty and inequality
 - each row identifies a unique combination of *country_code*, *year*, *reporting_level*, and *welfare_type*
 - *reporting_level*: Mostly national, but can be urban or rural
 - *welfare_type*: Income or consumption
- **pip wb:** regional/global estimates of poverty and inequality
 - each row identifies a unique combination of *region_code* and *year*.

Country-level estimates

```
1 set linesize 200
2 pip cl, clear
```

first observation

```
+-----+
| country_code  year  poverty_line  headcount    mean    median  welfare_type |
+-----+
|              AGO   2000           3.00      0.2695   8.3803   5.3003802   consumption |
+-----+
```

[Click here to display how to cite](#)

Variables

1 describe

Contains data from C:\Users\wb535623\AppData\Local\Temp\ST_4f2c_000001.tmp

Observations: 2,504 WB poverty at country level (8 Jul 2025)

Variables: 43 8 Jul 2025 05:31
(_dta has notes)

Variable name	Storage type	Display format	Value label	Variable label
country_code	str3	%9s		country/economy code
country_name	str24	%24s		country/economy name
region_code	str3	%9s		region code
region_name	str27	%27s		region name
reporting_level	str8	%9s		reporting data level
year	int	%8.0g		year
welfare_time	double	%10.0g		time income or consumption refers to
welfare_type	byte	%11.0g	welfare_type	welfare measured by income or consumption
poverty_line	byte	%6.2f		poverty line in 2021 PPP US\$ (per capita per day)

Ghanaian survey estimates

```
1 pip, country(GHA) n2disp(7) clear
```

first 7 observations

country_code	year	poverty_line	headcount	mean	median	welfare_type
GHA	1987	3.00	0.7822	2.3270	1.8898523	consumption
GHA	1988	3.00	0.7744	2.3897	1.9086236	consumption
GHA	1991	3.00	0.8036	2.2220	1.6923785	consumption
GHA	1998	3.00	0.6535	2.9585	2.2008481	consumption
GHA	2005	3.00	0.6089	3.3877	2.46647	consumption
GHA	2012	3.00	0.4177	4.8226	3.5122672	consumption
GHA	2016	3.00	0.3903	5.0688	3.7226515	consumption

[Click here to display how to cite](#)

Ghanaian survey estimates

- The default poverty line is \$3.00 (per capita, per day, in 2021 PPP-adjusted dollars)
- The mean is expressed in the same currency

Ghanaian survey estimates with 2017 PPP

```
1 pip, country(GHA) n2disp(7) ppp_year(2017) clear
```

first 7 observations

country_code	year	poverty_line	headcount	mean	median	welfare_type
GHA	1987	2.15	0.5839	2.3365	1.8976071	consumption
GHA	1988	2.15	0.5766	2.3995	1.9164557	consumption
GHA	1991	2.15	0.6417	2.2311	1.6993231	consumption
GHA	1998	2.15	0.4865	2.9707	2.2098793	consumption
GHA	2005	2.15	0.4247	3.4015	2.4765912	consumption
GHA	2012	2.15	0.2569	4.8424	3.5266795	consumption
GHA	2016	2.15	0.2521	5.0895	3.7379272	consumption

[Click here to display how to cite](#)

Different poverty line

```
1 pip, country(LUX) povline(10) clear
```

first observation

```
+-----+
| country_code  year  poverty_line  headcount      mean      median  welfare_type |
|-----|
|           LUX  1985          10.00      0.0047    49.4201    43.384774    income |
+-----+
```

[Click here to display how to cite](#)

Multiple poverty lines

```
1 pip, country(LUX) year(2022) n2disp(4) povline(10 20 50 100) clear
```

first 4 observations

country_code	year	poverty_line	headcount	mean	median	welfare_type
LUX	2022	100.00	0.5921	103.2674	87.469192	income
LUX	2022	50.00	0.2031	103.2674	87.469192	income
LUX	2022	20.00	0.0130	103.2674	87.469192	income
LUX	2022	10.00	0.0028	103.2674	87.469192	income

[Click here to display how to cite](#)

Estimates across selected countries

```
1 pip, country(LUX GHA VNM) n2disp(3) year(2022) clear
```

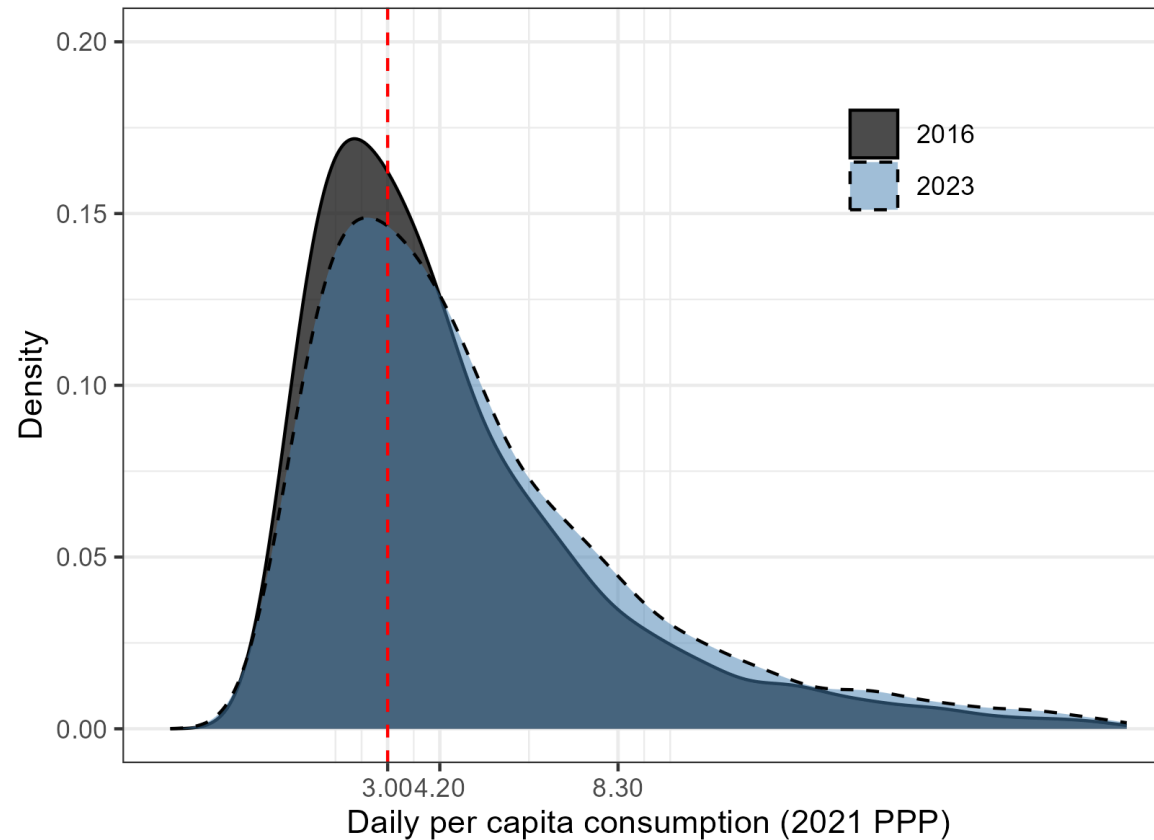
first 2 observations

country_code	year	poverty_line	headcount	mean	median	welfare_type
LUX	2022	3.00	0.0010	103.2674	87.469192	income
VNM	2022	3.00	0.0160	16.5866	13.319195	consumption

[Click here to display how to cite](#)

- Note that there is no survey for Ghana (GHA) in 2022 and hence not reported in the output.

Extrapolation for Ghana from 2016 to most recent year



Interpolated and extrapolated values

```
1 pip, country(GHA) year(2016/2025) n2disp(5) fillgaps nowcasts clear
```

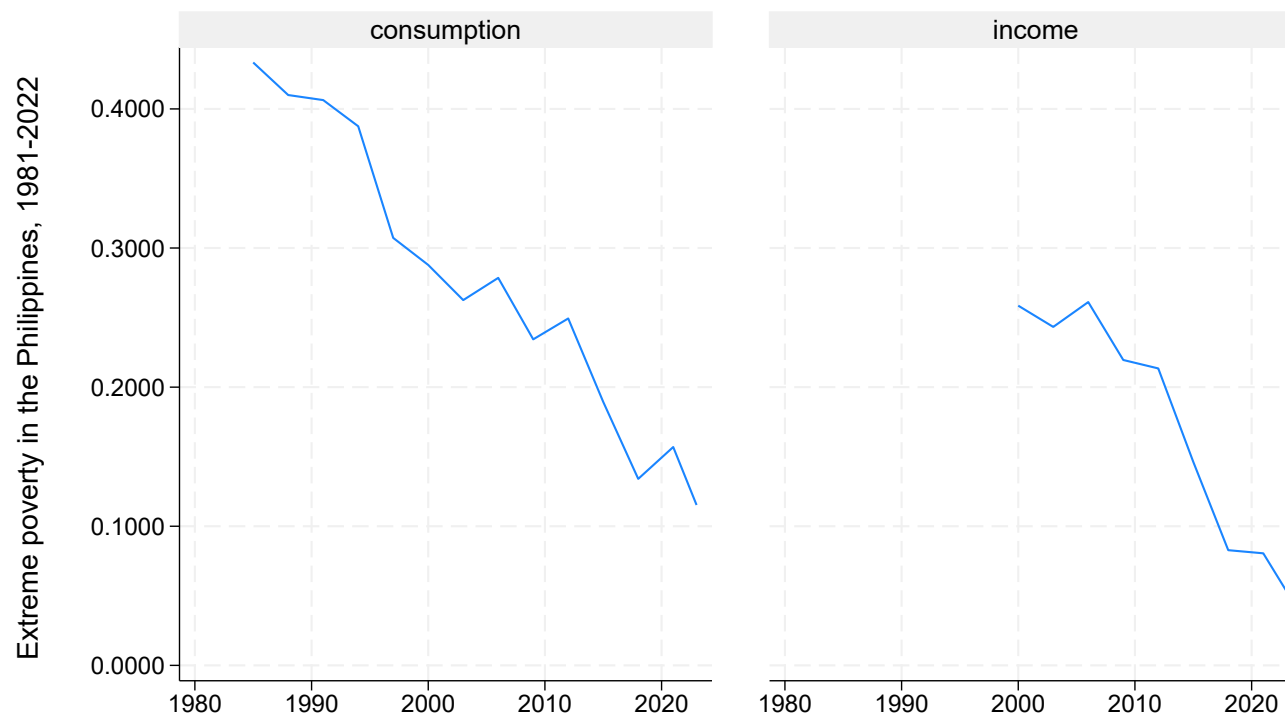
first 5 observations

country_code	year	poverty_line	headcount	mean	median	welfare_type
GHA	2016	3.00	0.4046	4.9053	3.5961474	consumption
GHA	2017	3.00	0.3852	5.1186	3.7610946	consumption
GHA	2018	3.00	0.3753	5.2626	3.8650692	consumption
GHA	2019	3.00	0.3593	5.4244	3.9859082	consumption
GHA	2020	3.00	0.3655	5.3686	3.9441622	consumption

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Poverty by the type of welfare aggregate

```
1 pip, country(PHL) clear
2 graph twoway line headcount year, by(welfare_type) xtitle("") ytitle("Extreme poverty in the Philippines, 1981-2022")
3 graph export "welfare_plot.svg", replace
```



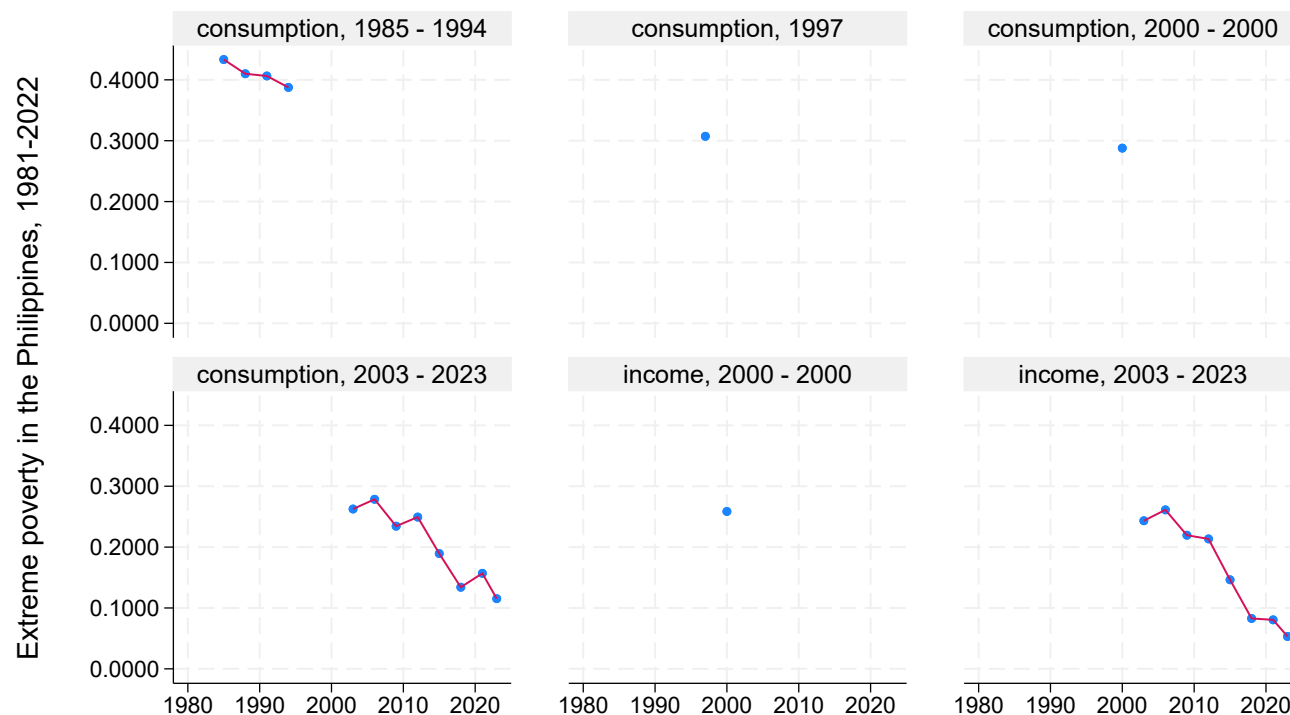
Graphs by welfare measured by income or consumption

Poverty by the type of welfare aggregate

```

1 pip, country(PHL) clear
2 graph twoway scatter headcount year || line headcount year, by(welfare_type comparable_spell, legend(off)) xtitle("") y
3 graph export "welfare_plot_2.svg", replace

```



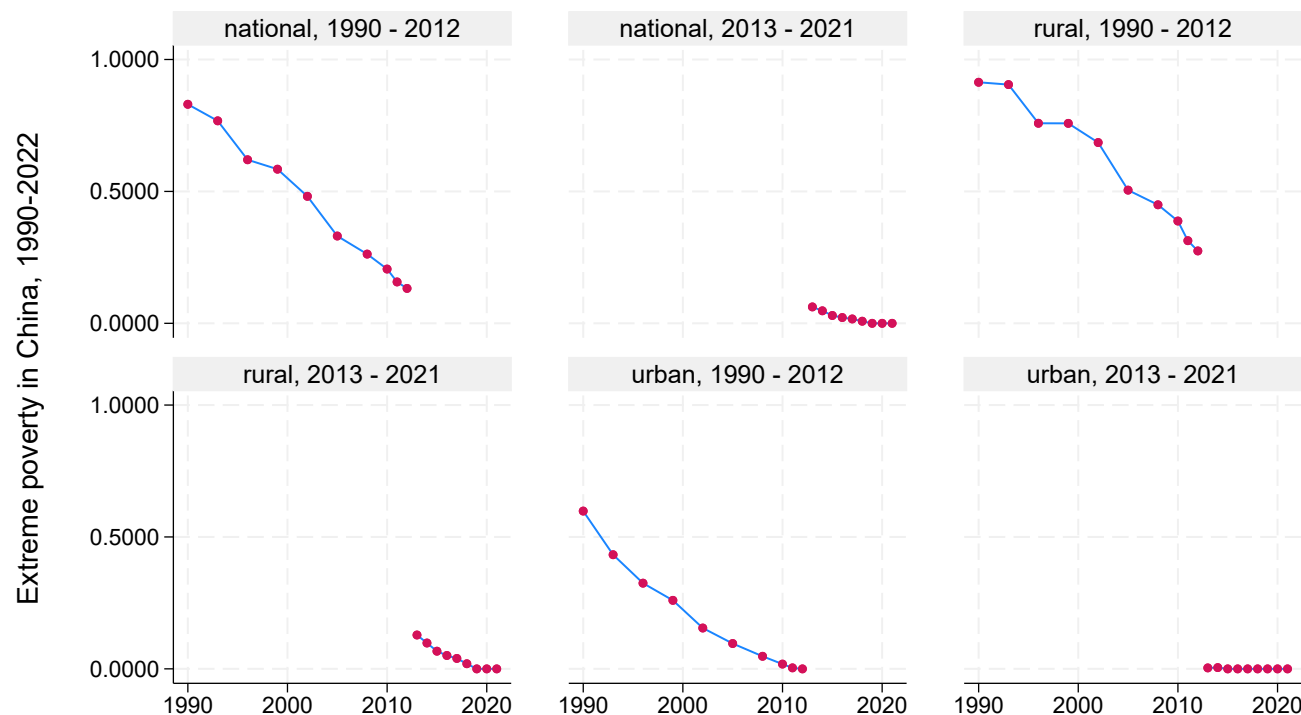
Graphs by welfare measured by income or consumption and comparability over time at country level

Poverty by reporting level

```

1 pip, country(CHN) year(1990/2022) clear
2 graph twoway line headcount year || scatter headcount year, by(reporting_level comparable_spell, legend(off)) xtitle("")
3 graph export "rep_level_plot.svg", replace

```



Graphs by reporting data level and comparability over time at country level

Within-country comparability

```
1 keep if reporting_level=="national"
2 list country_name reporting_level year headcount comparable_spell welfare_type
```

(38 observations deleted)

	count~me	report~l	year	headco~t	comparabl~l	welfare_~pe
1.	China	national	1990	0.8303	1990 - 2012	consumption
2.	China	national	1993	0.7676	1990 - 2012	consumption
3.	China	national	1996	0.6200	1990 - 2012	consumption
4.	China	national	1999	0.5843	1990 - 2012	consumption
5.	China	national	2002	0.4815	1990 - 2012	consumption
6.	China	national	2005	0.3310	1990 - 2012	consumption
7.	China	national	2008	0.2624	1990 - 2012	consumption
8.	China	national	2010	0.2059	1990 - 2012	consumption
9.	China	national	2011	0.1570	1990 - 2012	consumption
10.	China	national	2012	0.1324	1990 - 2012	consumption
11.	China	national	2013	0.0623	2013 - 2021	consumption
12.	China	national	2014	0.0474	2013 - 2021	consumption
13.	China	national	2015	0.0298	2013 - 2021	consumption

Data from LIS

```
1 pip, clear
2 keep if strpos(survey_acronym, "LIS")
```

```
1 tab country_name
```

country/economy name	Freq.	Percent	Cum.
-----+-----			
Australia	12	3.06	3.06
Austria	7	1.79	4.85
Belgium	6	1.53	6.38
Canada	44	11.22	17.60
Czechia	3	0.77	18.37
Denmark	4	1.02	19.39
Finland	4	1.02	20.41
France	12	3.06	23.47
Germany	30	7.65	31.12
Greece	2	0.51	31.63
Hungary	3	0.77	32.40
Ireland	6	1.53	33.93
Israel	25	6.38	40.31
Italy	17	4.34	44.64
Japan	13	3.32	47.96
Korea, Rep.	11	2.81	50.77
Luxembourg	18	4.59	55.36
Netherlands	5	1.28	56.63

Global and regional estimates

```
1 pip wb, clear
```

first observation

```
+-----+
| region_code  year  poverty_line  headcount    mean |
+-----+
|           WLD  1981           3.00    0.4735  11.8051 |
+-----+
```

[Click here to display how to cite](#)

```
1 levelsof region_code
```

There is a problem with PIP API server. Try again later

invalid syntax

r(197);

no variables defined

r(111);

r(111);

Global poverty, 1990-2022

Code

Plot

```
1 keep if year>=1990 & inlist(region_code,"WLD","EAP","SSA")
2 graph twoway line headcount year, by(region_name) ytitle("Poverty rate (%)") ///
3 ylab(,angle(horizontal))
4
5 graph export "pov_90_22.svg", replace
```

Many more functions

1 pip tables

Auxiliary tables available for :

01		aux_versions
02		countries
03		country_coverage
04		country_list
05		cpi
06		decomposition
07		dictionary
08		framework
09		gdp
10		incgrp_coverage
11		indicators
12		interpolated_means
13		metaregion
14		missing_data
15		national_poverty_lines
16		pce
17		pg_lnp
18		pg_svy
19		pop

4. Exercises

Exercise 1

Explore whether Kuznets' inverse-U hypothesis holds with data in PIP

1. Plot the relationship between an inequality metric and log mean welfare
2. Regress an inequality metric on a 2nd order polynomial of log mean welfare

Results

Code

Regression

Plot

```
1 pip, clear
2
3 gen logmean = log(mean)
4 gen logmean2 = log(mean)^2
5
6 graph twoway scatter gini logmean || ///
7 lpolyci gini logmean, ///
8 title("Kuznets curve?") ///
9 xtitle("Log welfare") ///
10 ytitle("Gini coefficient") ///
11 legend(off)
12
13 graph export "ex_1_plot.svg", replace
```

Exercise 2

COVID-19 impact on poverty and inequality

1. Calculate and plot the change in millions of extreme poor by region from 2019 to 2020
2. (Difficult) Calculate and plot the changes in Gini observed from 2019 to 2020 for countries with comparable data during these two years

Results (part 1)

Code

Plot

```

1 * Without the nowcast option, data will only be presented for regions with sufficient data coverage
2
3 pip wb, nowcasts clear
4 keep if year==2019 | year==2020
5 bysort region_code (year): gen changeinpoor = (pop_in_poverty-pop_in_poverty[_n-1])/10^6
6 keep if year==2020
7
8 graph hbar changeinpoor, over(region_code, sort(changeinpoor)) ytitle("Millions") title("Change in number of poor, 2019-2020")
9
10 graph export "ex_2a_plot.svg", replace

```

Results (part 2)

Code

Plot

```

1  pip, year(2019/2020) coverage(national)
2
3  *Only keep countries with comparable estimates in 2019 and 2020:
4
5  bysort country_code welfare_type comparable_spell: keep if _N==2
6  bysort country_code welfare_type comparable_spell (year): gen ginichange = gini-gini[_n-1]
7  keep if year==2020
8
9  *If countries have both income and consumption estimates, use only the latter:
10
11 bysort country_code: keep if _N==1 | welfare_type==1
12
13 *Something strange is going on with Kosovo:
14
15 drop if country_code=="XKX"
16
17 *Draw graph
18
19 graph bar ginichange, over(country_name, sort(ginichange) gap(0) lab(angle(90) labsize(small))) ///
20 title("Change in Gini index, 2019-2020") ytitle("Gini points")
21
22 graph export "ex_2b_plot.svg", replace

```

5. Additional resource

- [Methodological handbook](#): Describes all methodological details behind the estimates
- [Poverty & Inequality Briefs](#): 2-page summary of poverty and inequality in almost every developing country
- [Survey percentiles](#): 100 percentiles of each country-year survey distribution
- [Lined-up percentiles distribution](#): 1000 points on the global distribution
- [Accessing some of the microdata online](#)
- [More data and indicators through the user interface](#)
- [API access](#)

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Thank you for listening.

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